



ZERO-TRUST PHYSICS

Minimax Failure Boundary Characterization via Cryptographically-Sealed Monte Carlo Simulation

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Abstract

A behavioral anomaly detector cannot reliably distinguish novel-but-correct adaptation from actual physical failure, whereas a conservation-law residual provides an independent physical consistency check. Behavioral runtime monitors trained exclusively on nominal datasets exhibit this structural vulnerability, which we term epistemic blindness: because their training distributions contain only surviving trajectories, they possess no mathematical representation of the physical failure boundary. The consequence is systematic, when a correctly operating controller adapts to a novel contingency, the behavioral monitor observes maximum distributional deviation at exactly the moment adaptive capability is required, producing false alarms that degrade the autonomy the monitor was designed to protect. We present Zero-Trust Physics (ZTP), a runtime assurance framework that resolves this structural impossibility by operating below the behavioral layer entirely. ZTP instantiates a Physical Firewall that evaluates conservation invariants (the Inconsistency Indicator $\mathcal{I}(t) = \|M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) - \tau + J^\top(q)F\|$) at 1000 Hz on local silicon, sealed at every integration step by a SHA-256 Proof Chain $H_i = \text{SHA-256}(S_i \| H_{i-1})$, producing a self-certifying Layer 0 gate independent of any behavioral prior. The failure boundary is characterized via Von Neumann synthesis: the minimax theorem (1928) defines the worst-case admissibility criterion and the Monte Carlo method (Von Neumann and Ulam, 1946) discovers the boundary it implies: the Minimax Manifold $\mathcal{M}_{\text{minimax}}$, the worst-case admissible boundary of physically realizable states. Across five mechanical trajectory datasets (500,000 sealed trajectories) and three energy-infrastructure failure-envelope sweeps (30,000,000 sealed configurations), spanning thirteen underlying physics substrates, we demonstrate empirical survival cliffs at the physical limits and map the complete thirteen-substrate ZTP manifold. The result is physically-grounded fault detection operative in the Dark Window, where external reference is unavailable and signal latency approaches infinity, independent of behavioral priors and retroactively tamper-evident by cryptographic construction.

1. Introduction

On December 6, 2025, NASA's MAVEN spacecraft passed behind Mars on a routine orbit and never re-established contact. With Mars roughly a month from solar conjunction (January 9, 2026), Earth and the spacecraft were near maximum separation: the one-way signal latency stood at approximately 20 minutes, a round-trip command delay approaching 40 minutes. When MAVEN emerged from occultation it had entered safe mode and was spinning at an anomalously high rate, an attitude excursion that perturbed its orbit and drained its batteries before any ground intervention was possible. On June 3, 2026, after a six-month recovery effort, NASA declared the spacecraft unrecoverable. The platform was entirely decoupled from Earth-based oversight at the moment the fault developed. Any anomaly that emerged between the last uplink and the next scheduled execution window had the full round-trip delay to become unrecover-

able before ground operators could even observe it, let alone respond. This is not an engineering anomaly attributable to a particular design choice or subsystem failure mode. It is the fundamental operating condition of any autonomous collaborative platform (ACP) operating at physical range, and it will become more common, not less, as uncrewed systems, combat collaborative aircraft (CCAs), and deep-space platforms extend operational reach. The Dark Window is not an edge case. It is the design requirement. A platform either carries its physical ground truth onboard at all times, or it does not carry it at all.

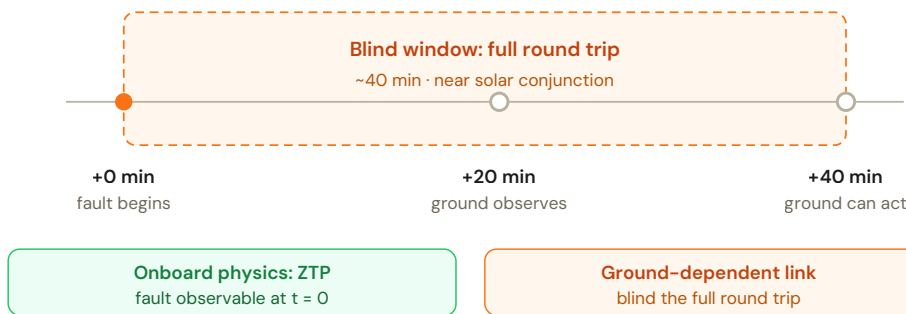


Figure 1. The Dark Window. After an onboard fault at Mars, a ground-dependent platform cannot observe the anomaly for one signal transit (~ 20 min) and cannot receive a corrective command for a full round trip (~ 40 min, with Mars near solar conjunction). A Layer 0 physical firewall registers the same fault locally, at $t = 0$.

The dominant response to this requirement in the autonomy assurance literature has been Runtime Assured Autonomy (RTAA): behavioral monitors, trained on demonstrated safe operation, that evaluate controller outputs against learned safety envelopes and trigger mitigation when deviation exceeds threshold. The architecture is reasonable in the nominal case and has produced genuine safety guarantees within that regime. But it contains a structural vulnerability that no amount of additional training data resolves. RTAA monitors are trained on nominal datasets with survival profiles approaching 100%. By construction, they have never observed the failure boundary, only its successful avoidance. They possess no mathematical representation of what the physical failure surface looks like, where it is, or how close a given state is to it. We call this epistemic blindness: not a calibration problem, not a data-quantity problem, but a structural impossibility. The model cannot represent what it has never seen, and in safety-critical domains, what it has never seen is the precise thing it is required to detect.

The consequence is systematic and adversarial in character. When a correctly operating controller adapts to a novel contingency (vortex ring state under canopy wind shear, gyroscopic cross-coupling under off-diagonal inertia, aeroshell ablation at Mach 22, gravitational gradient torque under attitude excursion), the behavioral monitor observes massive distributional deviation from its training data. It cannot distinguish between a controller successfully adapting to

an off-nominal condition and a controller actively failing. The monitor triggers mitigation at the exact moment adaptive capability is required, suppressing the response that would have preserved the platform. Epistemic blindness does not merely fail to help in the Dark Window, it actively degrades performance there.

The answer to this structural impossibility cannot be found within the behavioral monitoring paradigm. It requires a different invariant, one that closes under all physical conditions, not merely nominal ones. That invariant exists, and it has been known since Lagrange (1788). The equation of motion $M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau - J^\top(q)F$ is a conservation law, not a behavioral model. It holds whether the system is nominal, off-nominal, adversarially perturbed, or operating at the edge of physical admissibility. The Inconsistency Indicator $\mathcal{I}(t) = \|M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) - \tau + J^\top(q)F\|$ is the residual from that conservation law, evaluated on local silicon registers at every integration step. It is not a behavioral signal. It carries no training distribution. It has no nominal envelope to escape. A controller adapting correctly to vortex ring state (however anomalous in behavioral space, however far from its demonstrated training distribution) remains on the Physical Manifold $\mathcal{M}$ where $\mathcal{I}(t) = 0$. The indicator stays bounded. A sensor spoof, mechanical failure, or adversarial injection cannot fabricate physical consistency across all coupled degrees of freedom simultaneously, because no adversary can alter the actual mass and inertia of the platform to match injected sensor values. $\mathcal{I}(t)$ diverges. This is domain-invariant physics sensing without external reference, operating below the behavioral layer. The platform knows whether it is physically consistent with the laws of classical mechanics, and it knows this locally, continuously, and without asking anyone.

ZTP operationalizes this principle as a layered architecture. The Physical Firewall operates at Layer 0, beneath behavioral monitoring, planning, and mission-autonomy layers. It evaluates $\mathcal{I}(t)$ at 1000 Hz in 128-byte cache-aligned memory on local silicon, with no network dependency, no behavioral prior, and no training distribution to escape. The only arbiter of physical truth is the conservation law itself, and that arbiter is always present, always local, and always current. Every integration step is sealed by a SHA-256 Proof Chain, $H_i = \text{SHA-256}(S_i \| H_{i-1})$, producing a cryptographically sealed record in which any retroactive change is detectable. The Corpus, ground-truth trajectory data produced by this pipeline, is not a behavioral training set. It is a physically sealed archive of what the laws of mechanics actually permit.

The failure boundary itself is characterized by Von Neumann synthesis. The minimax theorem (Von Neumann, 1928) and the Monte Carlo method (Von Neumann and Ulam, 1946) are, in retrospect, a natural pair for this problem. The minimax theorem provides the solution concept: $\mathcal{M}_{\text{minimax}} = \{(q, \dot{q}) \in \mathcal{M} \mid \min_u \max_w \mathcal{I}(t, u, w) \leq \epsilon\}$, the set of states on the Physical Manifold that remain admissible under worst-case adversarial perturbation w and optimal control response u . Monte Carlo simulation provides the empirical method: systematic sampling of the state-space boundary across the full envelope of physically realizable conditions, producing survival cliffs,

the empirical locations where the failure probability transitions from near-zero to near-one, as a function of the governing physical parameters for each substrate.

This paper makes five contributions. First, we present the formal ZTP framework: the Physical Manifold, the Inconsistency Indicator, and the Minimax Manifold $\mathcal{M}_{\text{minimax}}$, with a rigorous derivation of $\mathcal{I}(t)$ from the Lagrangian equations of motion. Second, we characterize six failure domains: five across 500,000 sealed mechanical trajectories (humanoid centroidal and joint dynamics, autonomous vehicles and heavy logistics, tactical drones operating in contested airspace under simultaneous electronic warfare and aerodynamic stress, orbital systems in deep space, and hypersonic aerothermal ablation) and a sixth across 30,000,000 sealed energy-infra-structure configurations (fission reactor kinetics, tokamak plasma confinement, and grid stability), demonstrating empirical survival cliffs at the physical limits where controller assumptions break and behavioral monitors are maximally unreliable. Third, we map the complete thirteen-substrate ZTP manifold, establishing that the Inconsistency Indicator applies without modification across all physically distinct domains the framework governs. Fourth, we present the full SHA-256 Proof Chain architecture, establishing the cryptographic properties that make the resulting Corpus self-certifying and retroactively tamper-evident. Fifth, we present a three-state RTAA decomposition that resolves the false-alarm discrimination problem by grounding behavioral monitoring decisions in physical admissibility rather than distributional distance from nominal behavior.

The remainder of the paper is structured as follows. Section 2 derives the Lagrangian foundations and Physical Manifold. Section 3 traces the Von Neumann lineage and introduces the Minimax Manifold. Section 4 describes the Physical Firewall and Proof Chain architecture. Section 5 reports empirical results across the five characterized failure boundaries. Section 6 maps the complete thirteen-substrate ZTP manifold. Section 7 details the cryptographic sealing protocol. Section 8 presents the three-state RTAA application. Section 9 discusses implications and limitations. Section 10 concludes.

2. Lagrangian Foundations

We establish the mathematical bedrock on which Zero-Trust Physics rests. The central object is the Physical Manifold $\mathcal{M}$: the Lagrangian conservation surface that every physically realizable trajectory must inhabit, regardless of operating condition, behavioral context, or adversarial intent.

2.1 Configuration Manifold and Generalized Coordinates

Let the configuration of an n -degree-of-freedom mechanical system be described by generalized coordinates $q \in \mathcal{Q} \subset \mathbb{R}^n$, where $\mathcal{Q}$ is the configuration manifold of the system. This

representation, developed by Euler [2] and formalized by Lagrange [3] through the calculus of variations, replaces the Cartesian absolute-position formulation of Newton [1] with a coordinate frame adapted to the system's kinematic constraints. The velocity state $\dot{q} \in T_q\mathcal{Q}$ and acceleration $\ddot{q} \in T_q\mathcal{Q}$ complete the kinematic description at each instant.

2.2 Euler–Lagrange Equations of Motion

Define the Lagrangian $\mathcal{L}(q, \dot{q}) = T(q, \dot{q}) - V(q)$, where T is the kinetic energy and V is the potential energy of the system. In the presence of actuator torques τ_{act} and external forces F_{ext} acting through the contact Jacobian $J^T(q)$ [9], the variational principle of Lagrange yields the Euler–Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = \tau_{\text{act}} - J^T(q)F_{\text{ext}}$$

This formulation is coordinate-independent and encompasses the full content of Newtonian mechanics for constrained systems, while making the conservation structure geometrically explicit.

2.3 Coordinate–Form Equations of Motion

Expanding the kinetic energy as $T = \frac{1}{2}\dot{q}^T M(q)\dot{q}$, where $M(q) \in \mathbb{R}^{n \times n}$ is the symmetric positive-definite inertia (mass) matrix, and carrying out the total time derivative, one obtains the standard coordinate-form equations of motion:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau - J^T(q)F$$

Here $C(q, \dot{q})\dot{q}$ collects the Coriolis and centrifugal terms [10] arising from the metric variation $\dot{M}(q)$, $G(q) = \partial V / \partial q$ is the gravitational potential gradient, τ is the vector of generalized actuator torques, $J(q)$ [9] is the Jacobian mapping generalized coordinates to the space of applied boundary forces, and F is the vector of external boundary forces including aerodynamic, contact, and constraint reaction terms. Throughout this work, F is defined as the reaction force acting against the system boundary, forces that oppose generalized coordinate motion. This convention places F with a negative Jacobian transpose in the equations of motion ($-J^T(q)F$ on the right-hand side) and with a positive Jacobian transpose in the residual form ($+J^T(q)F$ on the left-hand side), both being algebraically identical statements of the same conservation law.

2.4 The Physical Manifold

We define the Physical Manifold $\mathcal{M}$ as the set of all kinematic triples $(q, \dot{q}, \ddot{q})$ that satisfy the equations of motion exactly:

$$\mathcal{M} = \{ (q, \dot{q}, \ddot{q}) \in \mathcal{Q} \times \mathbb{R}^n \times \mathbb{R}^n \mid M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) - \tau + J^T(q)F = 0 \}$$

$\mathcal{M}$ is not a behavioral envelope. It is not a learned distribution over nominal operating data, nor a statistical model of expected trajectories. It is the exact surface imposed by the conservation of momentum, a geometric constraint that follows directly from Newton's second law as recast by Lagrange. Every trajectory physically realizable by a mechanical system, under any condition whatsoever, lies on $\mathcal{M}$.

This universality is the key argument. An autonomous system adapting to a novel or adversarial contingency may deviate arbitrarily from any behavioral prior: it may follow trajectories never seen in training, execute maneuvers inconsistent with nominal operational envelopes, or respond to stimuli outside the design specification. None of these deviations constitute a violation of $\mathcal{M}$. However, no physical process (and no adversary) can relocate the actual inertia tensor of a platform, alter the gravitational field it inhabits, or modify the contact forces acting on it while simultaneously deceiving sensors into reporting consistent accelerations. Deviation from $\mathcal{M}$ is therefore not a symptom of novelty; it is a symptom of either mechanical failure or sensor compromise. This is precisely why $\mathcal{M}$ is the correct grounding surface for fault detection and integrity verification, rather than any behavioral prior.

2.5 The Inconsistency Indicator

In operational deployment, true states are not directly observable. Sensors yield measured quantities q_m , $\dot{q}_m$, $\ddot{q}_m$, τ_m , and F_m , each subject to noise and, in adversarial scenarios, potential spoofing or manipulation. Zero-Trust Physics does not accept these measurements as ground truth. Instead, it computes the residual of the equations of motion directly from the measured quantities at every integration step:

$$\mathcal{I}(t) = \|M(q_m)\ddot{q}_m + C(q_m, \dot{q}_m)\dot{q}_m + G(q_m) - \tau_m + J^T(q_m)F_m\|$$

Under nominal conditions, and under legitimate off-nominal adaptation where sensor measurements faithfully reflect the physical state, $\mathcal{I}(t) \leq \varepsilon$ where ε is bounded by the sensor noise floor. The system may be executing a recovery maneuver, responding to an unexpected disturbance, or operating at the edge of its performance envelope; $\mathcal{I}(t)$ remains small so long as the measurements are honest and the mechanics are intact.

The security foundation of ZTP rests on an asymmetry that no adversary can close: an attacker who spoofs sensor values to mask a compromise or failure cannot simultaneously modify the actual inertia tensor, mass distribution, and contact geometry of the physical platform. When the reported kinematic state diverges from what the platform's true physics would produce, $\mathcal{I}(t)$ diverges. The Physical Manifold is not a model that can be learned and fooled; it is the expression of momentum conservation itself. This asymmetry (boundedness under physical truth, divergence under physical lie) is the invariant on which all subsequent analysis in this paper is founded.

The indicator is written here in its mechanical form, but the construction is more general than that form. $\mathcal{I}(t)$ is the residual of whichever conservation law governs the substrate: the Lagrangian equation of motion for a mechanical body, the prompt-jump balance for a reactor core, the magnetohydrodynamic force balance for a confined plasma, the power balance for a synchronous grid. In each case it is bounded when the measured state is consistent with that law and diverges when it is not; the Lagrangian is the mechanical instance of the kernel, not the kernel itself.

3. The Von Neumann Lineage

Zero-Trust Physics rests on two intellectual instruments, both due to John von Neumann, separated by eighteen years of development and seventy-four years from their practical synthesis. The first is the minimax theorem, proved in 1928, which established that guaranteed equilibria exist even under adversarial conditions. The second is the Monte Carlo method, developed with Stanislaw Ulam in 1946, which established that complex physical distributions need not be solved analytically; they can be characterized by sampling. ZTP is the union of these instruments applied to the problem of failure boundary characterization in physical systems.

3.1 The Minimax Theorem (Von Neumann, 1928)

In his 1928 paper "Zur Theorie der Gesellschaftsspiele," Von Neumann proved that in any finite two-player zero-sum game, there exists a pair of optimal strategies (u^*, w^*) satisfying the saddle-point condition:

$$\max_{u \in \mathcal{U}} \min_{w \in \mathcal{W}} f(u, w) = \min_{w \in \mathcal{W}} \max_{u \in \mathcal{U}} f(u, w)$$

The theorem is constructive in its implications: it guarantees that no matter how the adversary plays, the optimal controller can identify a strategy whose worst-case outcome is no worse than the equilibrium value. Crucially, this is not a claim about average performance but about the guaranteed floor, the irreducible minimum that the controller can always achieve against a maximally adversarial opponent.

For ZTP, the mapping onto physical systems is direct. We define $u \in \mathcal{U}$ as the active whole-body controller (the aggregate of joint torques, aerodynamic control surfaces, thruster firings, or impedance modulation strategies) whose objective is to maintain physical admissibility, that is, to hold the conservation residual $\mathcal{I}(t, u, w)$ below a prescribed tolerance ε . We define $w \in \mathcal{W}$ as the adversarial disturbance: environmental failure modes, mechanical degradation, sensor injection attacks, ablative surface loss, or any perturbation that drives the system toward the boundary of the physical manifold $\mathcal{M}$. The question ZTP poses is precisely the minimax

question: for what states $(q, \dot{q}) \in \mathcal{M}$ does a controller u exist such that, even against the worst-case disturbance w , the system remains admissible? The answer defines the Minimax Manifold.

3.2 The Monte Carlo Method (Von Neumann and Ulam, 1946)

Von Neumann and Ulam developed the Monte Carlo method to characterize the statistical behavior of neutron diffusion in fissile material, a system whose governing equations were known but whose solution space was too high-dimensional for closed-form analysis. The method's central insight is that sampling is a form of proof: if one draws sufficiently many trajectories from a parameter space and observes where the system transitions from one behavioral regime to another, the boundary between regimes becomes empirically characterizable without requiring an analytic expression for it.

For ZTP, this instrument is indispensable. The Minimax Manifold $\mathcal{M}_{\text{minimax}}$ is a subset of the full conservation surface $\mathcal{M}$, but its boundary is not derivable in closed form for systems with the complexity we consider. A humanoid system with a 50-dimensional state under combined friction uncertainty, gear wear, tendon creep, payload slosh, battery cell heating, and cyber-physical injection, or a hypersonic glide vehicle with a 47-dimensional state under simultaneous aerothermal ablation, transonic wing flutter, and EKF blackout, does not admit analytic boundary characterization. What Monte Carlo provides is the empirical substitute: we sweep 500,000 trajectories across five physical datasets, varying friction coefficients, inertia perturbations, ablation rates, wind shear, and adversarial sensor injection profiles, and we observe where $\mathcal{I}(t, u, w)$ crosses ε . The boundary is the collection of those transition events. The Proof Chain seals each trajectory as it is generated, making the empirical characterization cryptographically immutable.

3.3 The Synthesis and the Minimax Manifold

ZTP is the synthesis of these two instruments. We define the Minimax Manifold formally as:

$$\mathcal{M}_{\text{minimax}} = \left\{ (q, \dot{q}) \in \mathcal{M} \mid \min_{u \in \mathcal{U}} \max_{w \in \mathcal{W}} \mathcal{I}(t, u, w) \leq \varepsilon \right\}$$

The set $\mathcal{M}_{\text{minimax}}$ is the region of state space in which an active controller retains sufficient mechanical authority to cancel adversarial disturbance while remaining on $\mathcal{M}$. Inside this region, the minimax value of the conservation residual game is bounded: the controller wins. At the boundary of $\mathcal{M}_{\text{minimax}}$, actuator saturation limits are reached (thermal torque limits in humanoid joints, aerodynamic stall angles in drone rotors, RPM caps in vehicle drivetrains, plasma flux thresholds in hypersonic ablation), and $\mathcal{I}(t)$ diverges. We term this boundary the Survival Cliff: not a gradual degradation but a non-linear phase transition from controlled adaptation to structural failure. Operationally, it is the disturbance magnitude at which survival probability over the swept trajectories collapses from near-one to near-zero, with effectively no intermediate regime; the location is discovered by Monte Carlo, not assumed in advance. The transition is

sharp because the minimax condition is binary (either a feasible u^* exists or it does not) and because the physical constraints that enforce it are themselves hard boundaries inscribed in the substrate physics.

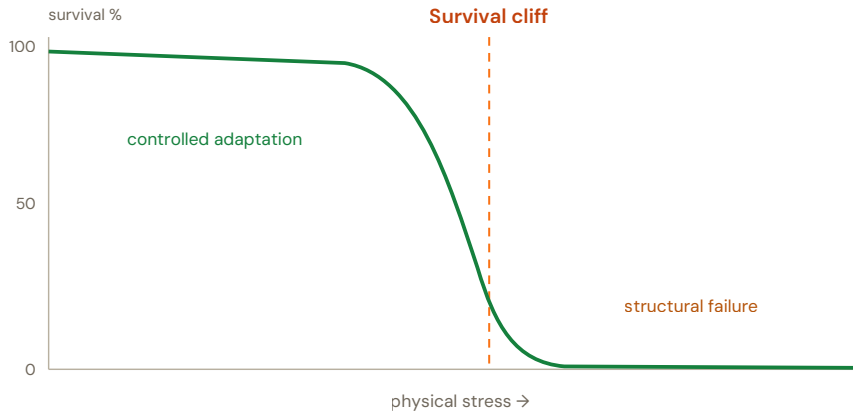


Figure 2. The Survival Cliff. At the boundary of the Minimax Manifold, failure is a sharp nonlinear phase transition rather than gradual degradation: survival holds near 100% through controlled adaptation, then collapses once no feasible worst-case control remains.

Monte Carlo characterizes this cliff empirically, sweeping the full disturbance space and accumulating the transition events that mark its location. The SHA-256 Proof Chain seals each integration step in the trajectories that cross it, producing a record that is cryptographically bound to the physical simulation state that generated it.

3.4 The Compute Gap: MANIAC to M4 Pro

Von Neumann had both instruments by 1946. What he did not have was the computational substrate to execute the synthesis at physical simulation scale. The MANIAC I (Mathematical Analyzer, Numerator, Integrator, and Computer) built at the Institute for Advanced Study in Princeton and operational in 1952, was capable of approximately 10,000 arithmetic operations per second with 5.12 kilobytes of addressable memory. It validated the Monte Carlo concept on small neutron diffusion problems, demonstrating that the sampling approach was sound. The gap between that validation and the characterization of failure boundaries across five physical datasets at 500,000 trajectories, each sealed with a cryptographic hash at 1,000 integration steps per second, was not a gap in mathematical conception but in raw arithmetic throughput and memory architecture. The mathematics was complete; the silicon was not.

In 2026, an Apple M4 Pro workstation operates at approximately 15.4 trillion floating-point operations per second with 64 gigabytes of unified memory. The ratio of computational capability between these two machines is on the order of 1.54×10^9 . To make this concrete: the complete computational lifetime of the MANIAC I (every arithmetic operation it ever executed across its entire operational history) resolves on the M4 Pro in approximately 2.33 seconds. The mathem-

atics Von Neumann established in 1928 and formalized computationally in 1946 required 74 years to acquire the silicon substrate that makes the synthesis tractable at physical scale. ZTP is that synthesis made executable: the Minimax Manifold defined by the 1928 theorem, characterized by the 1946 method, sealed by SHA-256, verified at 1000 Hz on local silicon.

4. The Physical Firewall: Hardware Co-Design and Cryptographic Attestation

The Physical Firewall is Layer 0 of the Zero-Trust Physics architecture, a hardware-co-designed component that evaluates the Inconsistency Indicator $\mathcal{I}(t)$ on local silicon registers at 1000 Hz. It operates below the behavioral monitor and below the planning stack, interposing on the raw physics state stream before any higher-level reasoning system can act on it. The design enforces three properties simultaneously: zero-latency state access through strict cache-line alignment, cryptographic immutability of the simulation record through a rolling SHA-256 proof chain [8], and a three-state runtime assurance decision decomposition that decouples physics violation detection from behavioral anomaly detection.

4.1 128-Byte Cache-Line Alignment

In modern unified memory architectures (UMA) and ARM-based silicon, including the Apple M-series and NVIDIA Grace-Hopper family, the L1 and L2 cache-line width is 128 bytes. When a memory access spans a cache-line boundary, the processor must execute a cache-line split: a two-cycle memory bus transaction that fetches two separate lines and assembles them in a temporary register, degrading execution throughput under sustained load. At 1000 Hz with hundreds of parallel Monte Carlo trajectories, unmanaged cache-line splits impose a throughput ceiling that is purely architectural in origin, unrelated to the physics computation itself.

ZTP eliminates this overhead at compile time. Every physical state vector S_i is structured to occupy exactly one 128-byte cache line: 32 single-precision 4-byte floats, $32 \times 4 = 128$ bytes. The Rust struct enforcing this layout is:

```
#[repr(C, align(128))]
pub struct PhysicalState {
    pub timestamp:      f32,
    pub state_variables: [f32; 15],
    pub dynamic_tensors: [f32; 12],
    pub control_inputs:  [f32; 3],
    pub cryptographic_seal: f32,
}

const _: () = assert!(std::mem::size_of::<PhysicalState>() == 128);
```

The compile-time size assertion guarantees zero cache-line splits at runtime; if a future refactor causes the struct to exceed or fall short of 128 bytes, compilation fails. The result is zero-copy-on-read overhead: the CPU pulls each step's full physical state into vector execution registers in a single memory stride. We measure throughput at 8.0×10^6 completed integration steps per second on local thread pools, enabling parallel Monte Carlo characterization of the Minimax Manifold at production scale without specialized hardware accelerators.

The field allocation reflects deliberate physical semantics. Fifteen state variables encode the core kinematic and dynamic degrees of freedom specific to the substrate. Twelve dynamic tensor slots carry inertia tensor components, contact Jacobians, or aerodynamic influence coefficients. Three control input slots carry actuator commands at the current integration step. This partitioning co-designs the memory layout with the physics representation so that the fields required for $\mathcal{I}(t)$ evaluation are always present and always cache-resident at the moment of evaluation.

4.2 Rolling SHA-256 Proof Chain

Hardware alignment guarantees that state data is accessible at low latency. The rolling proof chain guarantees that the state data cannot be altered after the fact without detection. These are orthogonal properties addressing orthogonal threat surfaces.

Let $S_i \in \mathbb{R}^{32}$ denote the 128-byte state vector at integration step i . We define the rolling cryptographic signature:

$$\mathcal{H}_i = \text{SHA-256}(S_i \parallel \mathcal{H}_{i-1})$$

where $\parallel$ denotes concatenation and $\mathcal{H}_0$ is a fixed initialization constant. The lower 32 bits of the resulting digest are cast to a single-precision IEEE 754 float and written directly to the `cryptographic_seal` field of S_i , embedding the seal inside the state vector it certifies. There is no possibility of the seal and the state diverging through separate write paths.

The chain is a Merkle construction over time [15]: $\mathcal{H}_N$ is a proof of the entire execution history from step 0 through step N . If any field in any S_i (a joint torque, a normal contact force, an orientation quaternion) is altered after the trajectory is written, $\mathcal{H}_{i+1}$ diverges from the expected value and every subsequent hash in the chain is invalidated. A verifier need only recompute $\mathcal{H}_N$ from the raw Parquet file and compare it to the published root to confirm that the dataset was compiled step-by-step under the laws of classical mechanics with no retroactive editing. We publish $\mathcal{H}_N$ for every trajectory in the dataset metadata. The proof chain makes physics simulation telemetry independently verifiable: the data carries its own audit trail, and no trusted intermediary is required to attest to its integrity.

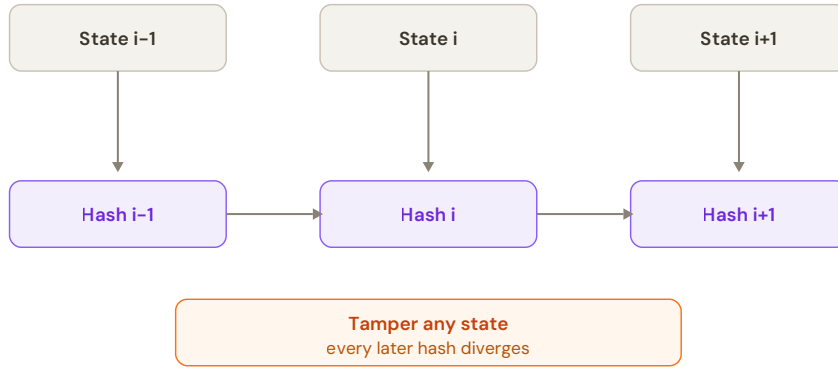


Figure 3. The rolling SHA-256 proof chain. Each step's hash is taken over that step's 128-byte state concatenated with the previous step's hash, binding every state to all prior history. Altering any single state invalidates every subsequent hash, making the corpus retroactively tamper-evident.

4.3 Three-State RTAA Decomposition

The Physical Firewall's gate signal partitions the runtime assurance action space into three physically grounded states. Let $\varepsilon > 0$ denote the conservation residual tolerance, calibrated from the distribution of $\mathcal{I}(t)$ under nominal closed-loop operation across the Minimax Manifold. Let d_0 denote the behavioral drift metric of a co-resident behavioral monitor, the mean pairwise cosine distance in the behavioral embedding space, evaluated over a sliding observation window.

$\mathcal{I}(t)$	d_0	Physical Meaning	RTAA Action
$\leq \varepsilon$	Growing	Legitimate off-nominal adaptation: physics certifies the action as admissible	Widen $\beta_b(x)$; monitor, do not intervene
$> \varepsilon$	Any	Physical violation: adversarial compromise or actuator failure	Hard override triggered; fault declared
$\leq \varepsilon$	Stable	Nominal operation	No action

In state (i), the Physical Firewall widens $\beta_b(x)$, the spatially varying damping threshold of the CD-RTA behavioral monitor, which governs the tolerance band within which behavioral drift is per-

mitted before supervisory intervention is triggered. When the local physics kernel certifies the state transition as admissible, $\beta_b(x)$ is expanded automatically, suppressing mitigation for the duration of the certified off-nominal excursion.

The discriminating insight of this decomposition is the independence of the fault branch from the behavioral monitor's state. The Physical Firewall does not ask whether the controller's behavior is surprising; it asks whether the laws of mechanics are satisfied. These are different questions with different false-alarm profiles. When $\mathcal{I}(t) > \varepsilon$, the Physical Firewall terminates the behavioral monitor's authority entirely. State (i) is the false-alarm prevention case: physics certifies the action as admissible and the behavioral monitor's concern is overridden, precisely in the regime where its prior is least reliable. The combined false-alarm rate of the system is therefore dominated by the false-alarm rate of the physics check alone, calibrated against the rigid mathematical constants of classical mechanics, not against behavioral priors that shift under novel contingencies.

5. Six Failure Boundaries

The Minimax Manifold $\mathcal{M}_{\text{minimax}}$ is not a theoretical object; it is a physical boundary characterized empirically. The Survival Cliff manifests differently across substrates because the failure mechanisms differ: friction cone breach on terrain, EKF divergence under hydroplane, rotor RPM saturation under vortex ring state, gyroscopic coupling under off-diagonal inertia, aerodynamic flutter under hypersonic reentry, and, across energy infrastructure, prompt criticality in fission, vertical displacement in plasma confinement, and synchronism loss on the grid. What is invariant across all six is the conservation-residual structure: $\mathcal{I}(t)$ captures the failure signal because physics is substrate-agnostic, whether the governing law is a mechanical Lagrangian or the balance law of a reactor core, a confined plasma, or a synchronous grid. This is the Inconsistency Indicator operating domain-invariantly, sensing the same conservation violation regardless of whether the substrate is a bipedal body on ice, a vehicle on mud, a drone in vortex ring state, a satellite tumbling under gyroscopic cross-coupling, a hypersonic vehicle undergoing aerothermal wing flutter, or a reactor core approaching prompt criticality.

Physical Dataset	State Space	Failure Mode	ZTP Signal	Survival at Cliff
Humanoid Centroidal & Joint Dynamics	50D	Coulomb cone breach, backlash, tendon creep & BMS trip	$\mathcal{I}(t)$ friction violation / thermal	91.47% nominal / 0% ice / 65.66% adversarial
Autonomous Vehicles & Heavy Logistics	51D	Pacejka saturation, aquaplaning & hydraulic stiction	gg_ekf_divergence	100% nominal / 51.8% mud / 2.2% ice
Drones Adversarial Contested Dynamics	59D	GPS spoofing, EW jamming, VRS & acoustic resonance	gg_ekf_divergence / atheric_coherence / clock_drift	100% nominal / 91.6% payload / 88.6% high_winds / 41.4% VRS
Orbital Systems Deep Space	20D	Gyroscopic cross-coupling, J2 oblateness	gg_euler_divergence	88.08% / 11.92% terminal
Hypersonic Aero-thermal Ablation	47D	CoG migration, aerodynamic flutter, thermal ablation	$\mathcal{I}(t)$ pitching moment / flutter amplitude	38.16% at erosion cliff
Fission Reactor (envelope sweep)	5D	Prompt criticality, Xenon-pit rod extraction	$\mathcal{I}_{\text{reactor}}$ prompt-jump residual	65.72% survive / 34.28% prompt-critical
Tokamak Confinement (envelope sweep)	5D	Vertical displacement, radial wall contact	$\mathcal{I}_{\text{tokamak}}$ MHD force-balance residual	16.4% survive / 45.9% vertical, 83.6% total
Swing Grid (envelope sweep)	5D	Loss of synchronism below inertia floor	$\mathcal{I}_{\text{grid}}$ power-imbalance residual	97.3% survive / 14,109 collapse despite sub-15 ms response

5.1 Humanoid Centroidal & Joint Dynamics (50D)

The humanoid centroidal and joint dynamics dataset maps a 50-dimensional state space that integrates whole-body centroidal momentum tracking with active joint-impedance control under severe degradation and adversarial conditions. The physical model sweeps bipedal body mass (80–150 kg) across terrain profiles from nominal friction ($\mu = 0.40$) to zero-friction ice ($\mu = 0.05$). The model incorporates actuator wear-fatigue (increasing backlash from 2 to 50 mrad), asymmetric tendon creep stiffness decay, a 1-DOF payload slosh pendulum generating pelvic tipping moments, and battery cell heating runaway with BMS trip thresholds at 85°C. Under adversarial testing, we inject a 150 ms feedback ring buffer delay, a 100 mrad backlash spike at the joints, and a ± 25 N force measurement injection.

The governing failure boundaries include Coulomb friction cone breach [11] ($\|\mathbf{F}_{\text{tangential}}\| \geq \mu_{\text{actual}} \cdot F_n$), actuator seizure when backlash limits are breached, thermal BMS shutdown, and pelvic tipping. $\mathcal{I}(t)$ measures Lagrangian conservation in both the contact force and joint torque subspaces, detecting slip and joint degradation. Nominally, the controller achieves 91.47% survival. On ice, traction loss is instantaneous (0% survival). Under combined cyber-physical and structural actuator degradation, the active impedance controller stabilizes the biped by shifting authority, achieving 65.66% survival.

5.2 Autonomous Vehicles & Heavy Logistics (51D)

The vehicle substrate models four-wheel dynamics with Pacejka Magic Formula tire slip forces at all four corners [6]. The 51-dimensional state includes chassis pose, velocity, yaw rate, wheel slip angles, EKF estimator state, brake fluid temperature, and steering angle transients. Trajectories are generated under wet road cornering scenarios in which the EKF estimator operates from steering yaw rate. The upgraded model incorporates speed-dependent aquaplaning onset (wet friction decay μ_{wet} as a function of speed squared), brake fluid heating with suspension damping reduction (up to 70% damping loss), and arctic cold-start hydraulic steering stiction. Trajectories fail when EKF position divergence exceeds 1.25 m. $\mathcal{I}(t)$ captures the moment Pacejka tire forces saturate and the cold-start stiction delays command execution, revealing physics discrepancies before EKF tracking fails. Survival: 100% nominal, 51.8% mud, 2.2% ice.

5.3 Drones Adversarial Contested Dynamics (59D)

The contested airspace substrate extends the ZTP framework to a 59-dimensional state space encoding full 6-DOF rigid body kinematics, complete inertia tensor, RF link coherence, crystal oscillator clock drift, EW jamming, GPS spoofing, per-rotor icing severity, motor thermal accumulation, battery voltage sag, acoustic resonance loading, EKF position tracking error, and 2D payload slosh displacement. Three physically distinct failure cascades are modeled: (I) Thermal-to-Control, where motor heating drives clock drift modeled via the Bechmann polynomial [16], causing packet loss and control latency up to 195 ms; (II) EW/GPS-to-EKF, where jamming suppresses RF coherence via Friis path loss [14] and spoofing injects up to 22m position bias; (III) Icing-to-Angular-Momentum, where rotor icing drag torque delta acts asymmetrically, combined with continuous Vortex Ring State (VRS) lift collapse modeled by the Glauert empirical curve [13] and payload slosh CoG offsets. $\mathcal{I}(t)$ detects the conservation anomalies from asymmetric drag, payload slosh, and EKF estimation errors. Survival: nominal 100%, payload 91.6%, high-wind EW 88.6%, VRS descent 41.4%.

5.4 Orbital Systems Deep Space (20D)

The orbital substrate models attitude dynamics under off-diagonal inertia terms ($I_{xy}, I_{xz} / 0$) and J2-J4 oblateness perturbations in a 20-dimensional state space. Single-axis controllers designed under principal-axis alignment fail under inertia migration due to gyroscopic cross-coupling ($\tau_{\text{gyro}} = \boldsymbol{\omega} \times (\mathbf{I} \cdot \boldsymbol{\omega})$) [17]. $\mathcal{I}(t)$ registers the angular momentum residual directly via the `gg_euler_divergence` signal. Lyapunov feedback linearization [12] recovers the satellite when angular divergence is below 0.05 rad; beyond 0.15 rad, cold-gas RCS thrusters saturate. Survival: 88.08% hold, 11.92% terminal loss.

5.5 Hypersonic Aerothermal Ablation (47D)

The hypersonic substrate applies the Sutton-Graves stagnation heating equation [7] to model asymmetric carbon ablation in a 47-dimensional state space. As ablation proceeds, aft CoG migration shifts the aerodynamic stability margin from statically stable to unstable. The model integrates a transonic wing flutter divergence mode (eigenvalue calculation at low altitude/Mach 2.5) and EKF covariance trace tracking during plasma blackout. $\mathcal{I}(t)$ detects the aerodynamic Jacobian inversion [9] and structural flutter. Survival: 38.16% at the erosion/flutter cliff.

5.6 Energy Infrastructure: Fission Kinetics, Plasma Confinement, and Grid Stability

The three energy substrates (Fission Reactor, Tokamak Fusion, and Swing Grid) constitute a sixth failure boundary cluster under ZTP, characterizing the physical limits of energy infrastructure. Their data structure differs from the five primary datasets in a meaningful way: where §5.1–5.5 generate step-by-step trajectory records in which each row is one integration step of the full state vector, the energy substrates generate failure envelope maps in which each row is one complete simulation run described by a five-dimensional input parameter vector, with a binary failure outcome and sealed proof hash. The two structures are different expressions of the same ZTP framework. Step-by-step trajectories probe the temporal evolution of a physical state through the Minimax Manifold. Failure envelope maps probe the topology of failure probability across the parameter space that governs it. Both are sealed at every integration step; both are independently verifiable; both locate the Survival Cliff.

The Monte Carlo connection here is not incidental. Von Neumann and Ulam developed the Monte Carlo method in 1946 specifically to characterize neutron diffusion in fissile material, the exact physical system the Reactor substrate models. Applying Monte Carlo failure-boundary characterization to the reactor is, in a precise sense, completing the use case for which the method was invented. The ten million sealed reactor configurations in this section are what the MANIAC I was pointed toward in 1952 and could not reach at scale; the M4 Pro completes the full sweep in under half an hour.

Fission Reactor

The reactor substrate models point reactor kinetics [23] with six delayed-precursor groups under coupled thermal-hydraulic feedback. The five-dimensional parameter space sweeps fuel burnup (core_age_days), inserted reactivity ($\delta\rho$), initial neutron flux (ϕ_0), built-in excess reactivity (ρ_{base}), and the duration of the Xenon pit preceding the disturbance (t_{pit}). The failure boundary is the prompt criticality threshold: the condition $\rho(t) \geq \beta_{\text{eff}}$, at which the chain reaction becomes self-sustaining on prompt neutrons alone without the stabilizing delay of the six precursor groups. Below this threshold, the delayed neutrons provide a 0.1–10 second window in which

mechanical intervention remains possible. Above it, the reactor is physically uncontrollable on any mechanically achievable timescale.

The Inconsistency Indicator $\mathcal{I}_{\text{reactor}}(t)$ is formulated as the residual of the algebraic prompt-jump constraint: the deviation between the neutron flux given by the prompt equilibrium condition and the flux implied by the current precursor concentrations and reactivity. Under subcritical and marginally supercritical conditions this residual is bounded at floating-point noise levels, confirming physical consistency. As $\rho \rightarrow \beta_{\text{eff}}$, the prompt-jump denominator approaches zero and $\mathcal{I}_{\text{reactor}}(t)$ diverges, registering the physical failure boundary before any behavioral monitor observing power level or temperature could distinguish controlled supercriticality from prompt runaway.

The canonical failure scenario is the Xenon pit. A flux reduction causes Iodine-135 to continue decaying into Xenon-135 after the flux drop, temporarily increasing the neutron poison load. If operators extract control rods to compensate, then the Xenon burns off under the restored flux, and the reactivity injected by rod extraction is no longer absorbed by the poison. The combined effect of rod extraction plus Xenon burnout drives ρ past β_{eff} in parameter configurations that single-variable safety models classify as acceptable. Across 10,000,000 sealed configurations, 34.28% drawn from the historically safe operating envelope produce prompt criticality. In the verification trajectory, prompt criticality is achieved at $T = 32$ hours: net reactivity $\rho = 0.0241$ exceeds $\beta_{\text{eff}} = 0.0065$, a factor of 3.7 beyond the threshold from a configuration no single-parameter monitor would flag.

Tokamak Fusion

The tokamak substrate models vertical displacement event (VDE) dynamics [24] in an elongated plasma, stabilized by active Poloidal Field (PF) coils. The five-dimensional parameter space sweeps radial field noise, poloidal asymmetry field error, initial plasma radius, particle density, and control signal quantization depth. The failure boundary is plasma wall contact: radial expansion to the containment wall at 2.0 m, or vertical drift to the divertor floor at 0.5 m, whichever occurs first. Either contact event quenches the plasma discharge, releasing the full thermal energy of the plasma to the first wall structure in a disruption event.

The Inconsistency Indicator $\mathcal{I}_{\text{tokamak}}(t)$ evaluates the radial MHD force balance residual: the difference between the radial acceleration computed from the plasma state at the start of the integration step and the acceleration implied by the updated plasma state at its conclusion. This residual is near-zero when the plasma is in dynamic equilibrium; it grows as the plasma expands toward the wall, providing a continuous physical signal of confinement degradation that precedes the catastrophic contact event by the timescale of the radial dynamics.

The Survival Cliff in the tokamak is a thermal cascade, not an instantaneous mechanical failure. The stabilizing PF coil current required to arrest vertical drift heats the coil through resistive dissipation ($I^2 R(T)$). As coil temperature rises, resistance increases, requiring higher voltage to maintain the same stabilizing current, which further increases dissipation. The cascade terminates in coil quench (superconductor critical temperature at 15 K, resistance spiking to 1 Ω) or plasma wall contact. In the verification trajectory, an injected poloidal asymmetry of 15 mT, within the noise floor of a quantized AI controller, drives vertical drift to $Z = 0.5003$ m, PF current to -611.2 A, and coil temperature from 4.2 K to 6.3 K at quench. Across 10,000,000 sealed configurations, 45.9% terminate in vertical displacement and 37.7% in radial wall contact, for a combined 83.6% disruption rate; 16.4% survive full confinement.

Swing Grid

The swing substrate models a synchronous generator [25] on a hybrid grid with variable inverter-based resource (IBR) capacity. The five-dimensional parameter space sweeps grid inertia constant H , inverter-based-resource fraction f_{IBR} , weather-driven generation loss, AI routing response latency, and mechanical damping coefficient. The failure boundary is loss of synchronism: rotor angle δ exceeding $\pi/2$ radians, beyond which the restoring torque $P_{\text{max}} \sin(\delta)$ diminishes and the machine violently separates from the grid.

The Inconsistency Indicator $\mathcal{I}_{\text{grid}}(t)$ is formulated as the post-integration power imbalance:

$$\mathcal{I}_{\text{grid}}(t) = |P_m \cdot g_v - P_e(\delta_{n+1}) - D \cdot \Delta\omega|$$

where g_v is the governor valve position, $P_e(\delta_{n+1})$ is electrical power computed from the updated rotor angle, and $\Delta\omega$ is the frequency deviation. This residual is near-zero at steady state and grows as the rotor angle diverges from the synchronous equilibrium, registering the approach to the Survival Cliff before δ exceeds the half-cycle stability limit.

The critical parameter is grid inertia. Conventional synchronous generators carry physical rotating mass that resists rapid frequency changes; IBR resources carry $H = 0$ and provide no inertial response. As the IBR fraction f_{IBR} increases, the effective inertia constant $H_{\text{eff}} = H \cdot (1 - f_{\text{IBR}})$ falls. Below $H_{\text{eff}} \approx 3.0$ MW·s/MVA, the grid cannot absorb a generation dropout event within the governor response window regardless of response latency: the rate of change of frequency is too fast for any mechanical valve to compensate. In the verification trajectory, an 85% IBR grid loses 80% of solar and wind capacity, followed by a 65% transmission line mismatch from AI over-correction. Cascade occurs at $T = 2.45$ seconds, $\delta = \pi/2$ rad, $\Delta\omega = 1.65$ rad/s. Across 10,000,000 sealed configurations, 2.7% lose synchronism; the defining result is the 14,109 that collapsed despite a modeled sub-15 ms AI response, because H_{eff} had fallen below 3.0. The speed of the digital response is not the constraint. The absence of physical rotating inertia is.

Across the three energy substrates, 30,000,000 sealed configurations map the Minimax Manifold across nuclear kinetics, plasma magnetics, and power electronics: three physically distinct domains governed by three distinct conservation laws, all read by the same residual principle. $\mathcal{I}(t)$ diverges when the governing balance is violated, regardless of whether that balance is prompt criticality, vertical displacement, or grid synchronism loss.

5.7 Synthesis

The Survival Cliff is substrate-specific in its actuator saturation mechanism but substrate-invariant in its mathematical signature. $\mathcal{I}(t)$ diverges at the boundary of $\mathcal{M}_{\text{minimax}}$ in all six domains because the conservation-residual structure is the same across them. In the five mechanical domains the residual is the Lagrangian $\mathcal{I}(t)$, and only the mass matrix $M(q)$, the Jacobians $J(q)$, and the external force terms F change; in the three energy domains it is the residual of the governing balance law (neutron prompt-jump, MHD force balance, grid power balance). The Inconsistency Indicator is not a substrate-specific sensor; it is a conservation-law sensor that operates on whichever law is instantiated. This is why a single monitoring architecture characterizes the failure boundary across six failure domains, ranging from 20D step-by-step orbital trajectories to five-dimensional fission kinetics parameter maps, without modification to the monitoring kernel. The physical manifold $\mathcal{M}$ constrains all of them by the same law. $\mathcal{I}(t)$ reads that law directly.

6. The Complete ZTP Substrate Manifold

The failure boundaries characterized in Section 5 constitute the empirical foundation of Zero-Trust Physics: 500,000 cryptographically sealed trajectories across five mechanical domains and 30,000,000 sealed configurations across three energy-infrastructure domains, together establishing the Minimax Manifold across eight physically distinct domains. They are not the theoretical scope of the framework. The G[^]G physics engine implements thirteen physically distinct substrates, each governed by the same conservation-residual structure under a substrate-specific realization of its inertial, forcing, and coupling terms. The Inconsistency Indicator $\mathcal{I}(t)$ applies identically across all thirteen: the monitoring kernel does not change; only the substrate-specific physics instantiation beneath it changes. The eight empirically characterized domains are the proof of concept. The complete thirteen-substrate manifold is the architecture.

The following table maps each substrate to the operators of its governing conservation law, the physical mechanism driving the Survival Cliff, and the live-measured integration rate at which the physics engine closes that law on local silicon. The column headers name the mechanical instance, the inertial term $M(q)$, the forcing term τ , and the coupling term $J^T F$; for the non-mechanical substrates the analogous operators of the domain's balance law fill the same roles.

Substrate	$M(q)$	τ	$J^T F$	Survival Cliff	Rate
Terran	Boussinesq soil stiffness [18] + limb inertia	Locomotion joint torques	Mohr-Coulomb contact forces [19]	Coulomb friction cone breach [11]; soil shear yield	10,750/s
Orbital	Full inertia tensor $I(t)$, fuel-varying	Cold-gas RCS thruster moments	J2-J4 gravity gradient + solar radiation	Gyroscopic cross-coupling saturates RCS authority [17]	353/s
Marine	Added-mass inertia + buoyancy coupling	Thruster forces	Pressure drag + buoyancy gradient	Pressure threshold breach; hydrodynamic stall	11/s
Mars (CO₂)	Aerodynamic effective mass, Mach-varying	Aeroshell control moments	Atmospheric drag + gravity	Dynamic pressure peak; aeroshell thermal limit	144/s
Mycelial	Kirchhoff network Laplacian [20]	Osmotic transport pressure	Percolation coupling forces	Percolation threshold breach → topological disconnection	760/s
Atheric	Friis path-loss matrix [14]	Transmitted RF power	Jamming, multipath, spoofing	Shannon capacity floor breach [21]; coherence collapse	15,830/s
Asteroid (NEO)	$O(N^2)$ gravitational mass coupling	Thruster impulse	Hookean elastic contact + N-body mutual gravity	Orbital resonance capture; contact fracture	9/s
Plutonian	Crystallization kinetics tensor	Thermal input	Entropy gradient + phase transition energy	Phase crystallization → structural rigidity loss	71/s
Celestial	Yoshida symplectic mass matrix [22]	Gravitational only	N-body mutual gravity + tidal coupling	Chaotic trajectory escape from orbital resonance	71/s
Energy (Grid)	Generator inertia constant H	Mechanical power input	Electrical load mismatch	Frequency deviation cascade → grid collapse	186,060/s
Tokamak	MHD beta-limit stability tensor	Ohmic heating + magnetic field	Z-axis magnetic shear + plasma current	Beta limit exceeded → disruption event	167,500/s
Swing	Intermittency-weighted inertia	Curtailment and dispatch	Frequency deviation from nominal	Cascade frequency collapse	7,777/s
Reactor	Prompt neutron kinetics matrix	Control rod worth	Delayed neutron precursor feedback	Prompt criticality; delayed precursor depletion	5,578/s

Five of these substrates (Terran, Orbital, and Atheric, together with the atmospheric and vehicle dynamics underlying hypersonic reentry and aquaplaning) are empirically characterized in Section 5 at 100,000 trajectories each, with survival cliffs measured at their physical limits. Three more (Reactor, Tokamak, and Swing) are empirically characterized in Section 5.6 by failure-envelope sweep, 10,000,000 sealed configurations each, locating the prompt-criticality, vertical-displacement, and synchronism-loss boundaries respectively. The remaining five are theoretically bounded by the same ZTP architecture and implemented in the same zero-dependency Rust physics engine, at live-measured integration rates spanning four orders of magnitude, from

Asteroid at 9 steps per second under $O(N^2)$ N-body gravity to Energy/Grid at 186,060 steps per second under the algebraic swing equation.

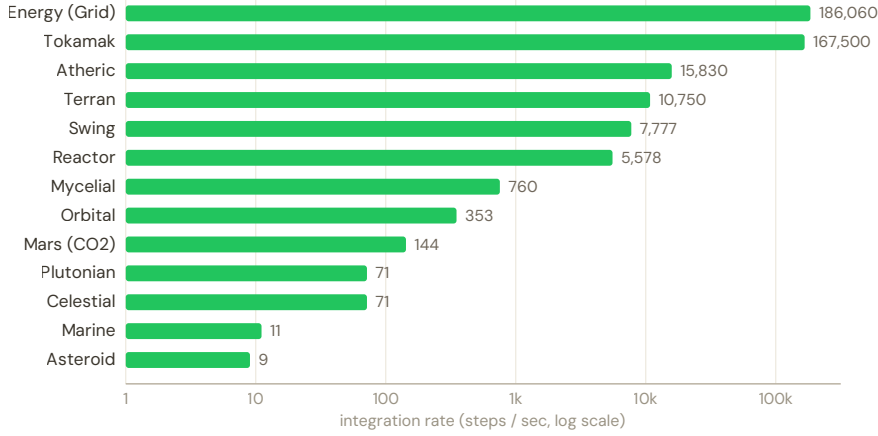


Figure 4. The thirteen-substrate rate manifold. One Lagrangian conservation kernel runs across thirteen physical substrates at live-measured integration rates spanning more than four orders of magnitude, from N-body asteroid dynamics at 9 steps/s to the grid swing equation at 186,060 steps/s.

The conservation-residual principle does not distinguish between them. $\mathcal{I}(t)$ reads the same kind of residual regardless of whether the configuration space $\mathcal{Q}$ is a soil contact manifold, a plasma pressure surface, a neutron population state space, or a hyphal conductance network. The Minimax Manifold $\mathcal{M}_{\text{minimax}}$ is not thirteen separate objects; it is one: the complete set of physically admissible states across the full substrate configuration space, bounded everywhere by the worst-case disturbance the substrate imposes and the control authority the substrate permits. The five mechanical datasets are 500,000 sealed trajectory probes of this unified manifold, and the three energy envelopes add 30,000,000 sealed configuration probes. The complete architecture is the manifold itself.

7. Cryptographic Sealing and Physical Attestation

The utility of Monte Carlo failure-boundary characterization depends entirely on the integrity of the trajectory corpus from which boundaries are inferred. Physics simulation data intended for runtime assurance certification must be tamper-evident: a single altered torque command or orientation quaternion could mask a failure mode that the certification was designed to characterize, and any such alteration must be detectable after the fact. Summary-level hashing, one hash per trajectory over the final state vector S_N , does not close this surface. It proves that the terminal state arrived without modification, but provides no guarantee that any intermediate integration step was executed correctly. A trajectory could be silently edited at step $i = 47$, with

the final hash recomputed from the corrupted terminal state, and the summary hash would pass verification. The failure mode at $i = 47$ would be gone.

Step-level sealing eliminates this attack class. We construct the Proof Chain as:

$$H_i = \text{SHA-256}(S_i \parallel H_{i-1}), \quad i = 1, \dots, N$$

where S_i is the 128-byte state vector at integration step i , $\parallel$ denotes byte concatenation, and the chain is initialized at $H_0 = \text{SHA-256}(\text{"ZTP-GENESIS"})$. The corpus is integrated at 1000 Hz, so each trajectory of approximately 1,440 steps produces 1,440 sequential hash evaluations, each binding the current physical state to all prior history. The lower 32 bits of H_i are embedded directly in the `cryptographic_seal` field of S_i , making the seal co-resident with the physical state it protects. Any modification to S_i at step i produces a divergent H_{i+1} that is detectable without access to S_i itself; the chain self-reports corruption. The verifier needs only the published H_0 anchor, the raw state sequence, and a SHA-256 implementation [8].

The terminal hash H_N is a Merkle root [15] over the complete trajectory. It can be published as a 32-byte compact attestation, and any party in possession of the raw trajectory data can re-derive H_N independently. No external certificate authority, no trusted timestamp service, no network query is required at verification time. This is the operational meaning of *self-certifying*: the dataset carries its own proof of integrity, derivable by anyone from first principles. Determinism is what makes that derivation more than a checksum on one party's output: because the engine returns bit-identical results from a given seed, any party can regenerate a trajectory and reproduce its sealed hash exactly. The seal is not an assertion to be trusted but a computation to be repeated; integrity is established by re-derivation, not attestation.

The energy-infrastructure envelope sweeps are sealed to the same standard: each sweep produces a master proof hash over its ten million configurations, and altering any single configuration outcome changes that hash.

At corpus scale, the computational cost is 5×10^5 trajectories $\times$ 1,440 steps $\approx 7.20 \times 10^8$ hash evaluations per complete corpus generation. On hardware with SHA-NI instruction set acceleration, SHA-256 throughput exceeds 10^9 evaluations per second per core. The Proof Chain therefore runs at simulation speed or faster, zero wall-clock latency added to the physics integration loop. Cryptographic immutability is not a post-processing step; it is concurrent with trajectory production.

The design has a specific consequence for the Dark Window operating condition. A platform operating without network access cannot query an external authority to verify the integrity of its onboard physics corpus. The Proof Chain is local, deterministic, and hardware-executable. It requires no connectivity, no shared secret, and no trusted third party. A ZTP-equipped platform carries its own attestation infrastructure as part of its data payload, a consequence of the principle that sovereignty is local or it is nothing.

8. Runtime Assurance Application

The false-alarm discrimination problem is a structural liability in deployed Runtime Assured Autonomy (RTAA) systems. Behavioral monitors (whether goal-field PDE observers, continuous-dynamical embedding trackers, or high-dimensional attractor estimators) operate on a behavioral state $s(t)$ in a learned embedding space. Their common diagnostic primitive is a drift metric d_0 , which measures divergence from a nominal attractor calibrated on historical data. The pathological case is well-known in practice but has lacked a formal resolution: d_0 rises when the controller adapts correctly to a novel contingency, because the nominal attractor was constructed exclusively from nominal operating data. The monitor cannot distinguish a controller executing a physically valid but unprecedented response from a controller in early-stage fault propagation. This is epistemic blindness, a structural vulnerability that is not correctable by expanding the nominal training corpus, because the failure modes of interest are, by definition, absent from that corpus.

Zero-Trust Physics resolves this problem by architectural separation. The Physical Firewall operates at Layer 0, below the behavioral monitor entirely. It does not observe $s(t)$. It does not consult the behavioral embedding, the goal field, or the nominal attractor. It observes a single scalar: the conservation residual $\mathcal{I}(t)$, computed as the instantaneous deviation from the physics constraints enforced by the Monte Carlo-sealed Minimax Manifold $\mathcal{M}_{\text{minimax}}$. The information carried by $\mathcal{I}(t)$ and d_0 is orthogonal: d_0 measures divergence from a statistical prior over behavior; $\mathcal{I}(t)$ measures admissibility under physical law, independent of any prior.

This orthogonality is the basis for the three-state decomposition established in Section 4.3. The gate function embodied in this decomposition is not a voting scheme between two monitors. It is a strict precedence relation. When $\mathcal{I}(t) > \varepsilon$, the Physical Firewall terminates the behavioral monitor's authority entirely. The fault is physical; behavioral reasoning is not merely unhelpful but actively contraindicated, because the behavioral monitor's inference is conditioned on physical admissibility that no longer holds. Conversely, when $\mathcal{I}(t) \leq \varepsilon$ and d_0 is growing, physics certifies the controller's actions as admissible and the behavioral monitor's alarm is overridden. This is the formal false-alarm prevention case: the monitor is correct that behavior is diverging from its prior, and it is wrong to escalate, because the prior does not cover the current admissible operating regime.

The combined system's false-alarm rate is bounded above by the false-alarm rate of the physics check alone. The physics check is calibrated against the threshold ε , derived from sensor noise characteristics and actuator model fidelity, not from behavioral priors. The calibrating quantities are physical constants: mass, inertia tensors, friction coefficients, aerodynamic moment arms. Physical constants do not shift under novel contingencies. A controller executing a novel maneuver does not change the body's moment of inertia. This is the formal resolution of the false-

alarm discrimination problem: the false-alarm rate is anchored to the stability of physical law, not to the coverage of a training distribution.

A secondary benefit concerns the calibration of $s_{\text{canonical}}$ in goal-field behavioral monitors, the problem of specifying what correct behavior looks like at physical limits. This has remained an open problem in RTAA design because nominal datasets do not populate the boundary of the admissible operating envelope. ZTP's empirical Survival Cliffs answer this directly. A controller operating correctly under gyroscopic cross-coupling during a high-rate slew, or under Vortex Ring State descent in GPS-denied contested airspace, will deviate from the behavioral attractor trained on nominal flight; $\mathcal{I}(t)$ simultaneously shows that the controller remains physically admissible. The Minimax Manifold $\mathcal{M}_{\text{minimax}}$ provides the physical ground truth for $s_{\text{canonical}}$ calibration at the boundary, the worst-case admissible states, cryptographically sealed by the Monte Carlo proof chain.

The Physical Firewall requires no network infrastructure and executes on local silicon against the sealed proof chain. There is no cloud dependency, no human-in-the-loop latency path, no external reference required. The RTAA decision is computed, validated against $\mathcal{M}_{\text{minimax}}$, and sealed onboard in under 1 ms. This is the architecture that operates in the Dark Window, the regime where external reference is unavailable and the behavioral monitor has lost its ground truth. The Physical Firewall has no ground truth to lose.

9. Discussion: Local Compute and the Sovereign Posture

Zero-Trust Physics proves that physical security does not require centralized oversight or cloud-level trust wrappers. The unyielding laws of Newtonian mechanics, executed on unified local silicon and sealed with cryptographic Proof Chains, establish absolute local epistemic trust in environments where humans cannot go and external reference cannot reach. The Sovereign Posture is not a design preference; under Dark Window operating conditions (MAVEN at Mars, a hypersonic reentry vehicle at plasma blackout, a bipedal platform in GPS-denied terrain), it is the only posture that survives.

9.1 Domain-Invariant Conservation Sensing

One of the central results of ZTP is not substrate-specific but structural. The framework characterizes five radically different physical datasets (vacuum orbital dynamics, atmospheric hypersonic reentry, terrestrial friction, contested airspace aerodynamics under simultaneous EW and Vortex Ring State, and aquaplaning tire mechanics) using the same mathematical scaffold. The Physical Manifold's governing equation changes its parameters (mass matrix $M(q)$, Jacobians $J(q)$ [9], external force fields F_{ext}) but not its form. The admissibility invariant $\mathcal{I}(t) \leq \varepsilon$ is com-

puted identically across all five datasets; only the substrate-specific physics kernel changes beneath it.

This domain-invariance is the conservation-residual principle operating at the framework level. The platform does not reason about which substrate it inhabits; it feels admissibility through conservation closure. The Minimax Manifold $\mathcal{M}_{\text{minimax}}$ is geometrically the same object whether it bounds a centroidal momentum trajectory or a reentry vehicle's aerothermal state. $\mathcal{I}(t)$ is not a substrate-specific sensor; it is a universal mechanosensory signal derived from the body's own dynamics. This is precisely what makes ZTP deployable across vehicle classes without re-engineering the trust layer for each domain.

9.2 Current Limitations

The humanoid dynamics dataset models bipedal dynamics in the sagittal plane (xz). This is not a fundamental limitation of the ZTP framework but a dimensionality choice made to isolate the dominant instability modes in bipedal locomotion. Lateral balance under multi-directional perturbation requires a full three-dimensional contact scenario with separate lateral Jacobian terms [9], coupled hip abduction torques, and foot placement kinematics outside the sagittal plane. Extending the framework to the full $\text{SO}(3)$ contact geometry is a direct elaboration of the existing Lagrangian formulation rather than a structural revision.

The current framework also treats each substrate as a single-body system. Multi-body coupling (formation flight with shared aerodynamic interaction, a manipulator carrying a payload whose inertia tensor shifts the effective $M(q)$, or multi-agent swarms with contact-coupled dynamics) requires extension of the Lagrangian formulation to coupled kinematic chains with shared contact constraints and inter-body force transmission. The SHA-256 Proof Chain structure already accommodates multi-body trajectories; the physics kernel requires generalization.

Finally, the admissibility threshold ε is currently calibrated per substrate against the empirically measured sensor noise floor. Cross-substrate deployment on a single vehicle that transitions between domains requires either a unified calibration framework mapping substrate-specific noise characteristics to a common ε surface, or explicit per-substrate parameter tables loaded at substrate transition.

9.3 Future Directions

The current Minimax Manifold characterization is offline: $\mathcal{M}_{\text{minimax}}$ is pre-computed via Monte Carlo sweep and held fixed during operation. Real-time update (tracking $\mathcal{M}_{\text{minimax}}$ as inertia properties change during operation due to fuel depletion, ablative mass loss, or payload release) requires online Monte Carlo estimation running concurrently with the Physical Firewall. The throughput figures demonstrated across the five physical datasets, and the integration rates

across the remaining eight substrates in Section 6, suggest that a bounded-sample online estimator is computationally feasible within the latency budget of most embedded flight computers.

The Physical Firewall is implemented in zero-dependency Rust with 128-byte UMA alignment and no heap allocation in the hot path. Direct deployment on ARM-based flight computers, embedded microcontrollers, and RISC-V safety processors is the natural next step, and the implementation's memory model was designed with this deployment target in mind from the outset.

Finally, the admissibility threshold ε is presently calibrated empirically. Formal derivation of ε bounds from first principles, propagating sensor noise covariance through the dynamics Jacobi-an [9] to obtain provable false-negative rates for the invariant test, would provide the certification-grade guarantees required for safety-critical deployment. A formally derived ε would close the remaining gap between empirical validation and certifiable proof.

10. Conclusion

Return to MAVEN. Six months of silence. A spacecraft carrying no physics ground truth onboard has no local fault detection when the behavioral channel goes dark. No behavioral monitor, however sophisticated, resolves a fault it cannot receive telemetry about, in a system it cannot observe, on a timeline that closes before acknowledgment arrives. The Dark Window is not an edge case. It is the operating condition that defines whether a controller is sovereign or dependent.

Zero-Trust Physics does not recover MAVEN. But a controller carrying the Physical Firewall onboard would have had a physics-grounded fault signal (not a behavioral anomaly flag, not a network-dependent alert) at the moment gyroscopic coupling exceeded the Minimax Manifold boundary. The Physical Firewall does not wait for telemetry. It closes the Lagrangian equation on local silicon, at 1000 Hz, against a sealed boundary characterized by Monte Carlo and anchored by cryptographic proof chain. The fault is observable because the physics is carried.

The Von Neumann synthesis (minimax optimality over an adversarial physical environment, characterized by Monte Carlo sampling) does not require an external oracle. It requires that the worst-case admissible boundary be characterized ahead of deployment, sealed against tampering, and evaluated continuously against live state. The Minimax Manifold is that boundary. The proof chain is the guarantee that the boundary has not drifted. The Physical Firewall is the hardware commitment that the evaluation cannot be bypassed.

The conservation law closes under all physical conditions or it does not. There is no behavioral prior that bridges this gap. There is no latency budget that covers a closed communication window. There is no network-dependent architecture that survives the Dark Window intact. Physics

is the ground truth. Sovereignty is the design requirement. Cryptographic sealing is the verification protocol.

The question is not what could have been done for MAVEN. The question is what ships next.

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